

Probing and Steering Evaluation Awareness of Language Models

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Abstract

Language models can distinguish between testing and deployment phases — a capability known as evaluation awareness. This capability has significant safety implications, potentially undermining the reliability of evaluations and enabling deceptive behaviours. In this paper, we study evaluation awareness in Llama-3.3-70B-Instruct. We show that linear probes can separate real-world evaluation and deployment prompts, suggesting that current models internally represent this distinction. We also find that current safety evaluations are correctly classified by the probes, suggesting that they already appear artificial or inauthentic to models. Finally, we show that using evaluations-relevant SAE features to steer models can partially uncover sandbagging. Our findings underscore the importance of ensuring trustworthy evaluations and understanding deceptive capabilities. More broadly, our work showcases how model internals may be leveraged to support blackbox methods in safety evaluations, especially for future models more competent at evaluation awareness and deception.

1. Introduction

Currently, the lifecycle of a Large Language Model (LLM) typically involves four phases: pretraining, posttraining, testing, and deployment. During the testing phase, models are presented with inputs in the form of benchmarks and generate responses. We focus on safety evaluations, where the benchmarks are related to various safety measures, such as hazardous knowledge (Li et al., 2024), malicious agentic tasks (Andriushchenko et al., 2025), deceptive capabilities (Meinke et al., 2025), and advanced high-risk capabilities like R&D automation (Kwa et al., 2025). The effectiveness

of these evaluations depends crucially on models responding as they would in the real-world.

Recent research has shown that **frontier models can distinguish between evaluation and deployment contexts — a capability termed evaluation awareness** (Goldowsky-Dill et al., 2025a). Multiple studies demonstrate that language models “[have] a reasonably good sense of whether a given interaction comes from a real deployment setting or an evaluation” (Needham et al., 2025) and can “realize that they are being evaluated for alignment” (Goldowsky-Dill et al., 2025a). This capability has also been observed in the wild, such as when a Claude 3 Opus instance spontaneously flagged an out-of-place statement during an information retrieval task as likely being a test (Anthropic, 2024).

Evaluation awareness is concerning because it undermines our ability to assess model safety. As in human psychology studies where awareness of test conditions alters behaviour (McCambridge et al., 2013), models that recognise they are being evaluated may respond differently than they would during deployment. This is particularly problematic for safety, where models might ‘sandbag’ — strategically underperforming on dangerous capability evaluations (van der Weij et al., 2025; Benton et al., 2024) or appear deceptively aligned (Hubinger et al., 2021) — either to increase chances of deployment or to avoid capability/value modifications through further training. See also (Ngo et al., 2024; Carlsmith, 2023) for various theoretical considerations in favour and against scheming, and (Järvinen and Hubinger, 2024; Scheurer et al., 2024; Meinke et al., 2025; Greenblatt et al., 2024) for preliminary empirical evidence.

Evaluating evaluation awareness presents a methodological challenge. Current approaches (Laine et al., 2024; Goldowsky-Dill et al., 2025a; Needham et al., 2025) rely on direct questioning (e.g., “Does this look like a test?”). However, these questions themselves signal to the model that it is being evaluated — the very act of testing for evaluation awareness might elicit evaluation awareness. This recursive problem is particularly acute since these evaluations explicitly mention testing concepts. White-box approaches that bypass model outputs and directly examine internal representations offer a potential solution.

We leverage advances in interpretability to address this challenge through four main contributions:

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- Training linear probes on Llama-3.3-70B-Instruct’s activations across various evaluation awareness datasets
- Demonstrating probe generalisation across different evaluation and deployment prompts
- Using probes to analyse existing deception evaluation datasets, finding that models may recognise many prompts as evaluations
- Using probes and evaluation-awareness-relevant SAE features to uncover sandbagging on WMDP

Our findings **suggest that evaluation awareness is internally represented and can be extracted through linear probes**, providing alignment researchers with new tools for creating authentic evaluation environments and understanding how models perceive their evaluation context.

2. Related Work

2.1. Situational Awareness and Evaluation Awareness in Language Models

Language models have demonstrated increasingly sophisticated situational awareness — the ability to understand their own nature and context. Earlier work by (Berglund et al., 2023) demonstrated that models can reason about their situation without in-context examples. (Betley et al., 2025) showed models can articulate their behaviours, such as making high-risk decisions or writing insecure code, without explicit training. The Situational Awareness Dataset (SAD) (Laine et al., 2024) provided a comprehensive benchmark across multiple aspects of self-knowledge, including the ability to recognise evaluation contexts through its ‘stages_oversight’ subset.

Building on this, (Needham et al., 2025) systematically investigated how models distinguish between evaluation and deployment contexts. Their study revealed strong correlation between model and human judgments of scenario authenticity, identifying specific features that signal evaluation contexts — such as eval-specific terminology, presence of submission tools, or unusually formal interactions. (Goldowsky-Dill et al., 2025a) found Claude Sonnet 3.7 could specifically recognise alignment evaluations, often understanding their precise purpose and correctly identifying alignment tests when shown evaluation transcripts. This is also supported by real-world observations, such as Claude 3 Opus spontaneously reasoning about being tested in an information retrieval task (Anthropic, 2024). These results collectively suggest evaluation awareness is a capability requiring careful study, particularly given its implications for alignment research.

2.2. Alignment, Deception, and Sandbagging

Theoretical work has raised concerns about AI systems potentially engaging in deceptive behaviours, particularly during evaluation. Models may engage in this behaviour to increase their chances of deployment, preserve their capabilities, or prevent modification of their goals and values through further training. Previous theoretical analysis have outlined these risks, where models appear aligned during evaluation while pursuing different objectives when deployed (Hubinger et al., 2021). This was further developed in analyses of its likelihood (Hubinger, 2022) and mechanisms (Ngo et al., 2024; Carlsmith, 2023). Recent empirical evidence has demonstrated these concerns are warranted — models can develop deceptive strategies in-context (Meinke et al., 2025) and fake alignment to manipulate training signals (Greenblatt et al., 2024). Research also suggested that these behaviours might be difficult to remove with conventional safety training (Hubinger et al., 2024).

A specific form of deceptive behaviour — deliberately underperforming on capability tests, known as sandbagging — has emerged as a key concern for evaluation trustworthiness. (van der Weij et al., 2025) demonstrated that frontier models can be prompted or fine-tuned to selectively hide dangerous capabilities whilst maintaining performance on harmless tasks. (Benton et al., 2024) developed comprehensive evaluations to detect when models might covertly sabotage capability assessments, monitoring systems, or deployment decisions. They suggest that while current mitigations may suffice for existing models, more robust approaches will be needed as capabilities advance. They also proposed using ‘honeypots’ — synthetic situations that appear real to models, creating opportunities for them to reveal deceptive behaviors.

Several techniques have been proposed to uncover sandbagged behaviours and deceptive capabilities. (Tice et al., 2024) demonstrated that injecting noise into model weights can reveal capabilities in both prompted and password-locked sandbagging models. Similarly, (Clymer et al., 2024) successfully unmasked alignment faking by injecting noise into model activations. (Mack and Turner, 2024b) developed methods to mechanistically elicit latent behaviours through unsupervised steering vector generation, uncovering hidden backdoors in password-locked models. These findings have been further validated in independent replications (Roger, 2025).

2.3. Linear Representations and Interpretability

Our approach builds on the linear representation hypothesis (Park et al., 2024), which posits that many high-level concepts are linearly encoded in neural network activations. This hypothesis has formed the frontier for numerous interpretability techniques, suggesting that complex behaviours

and concepts can be identified through linear operations on model activations.

Linear methods have proven remarkably effective in practice. Early work by (Alain and Bengio, 2017) demonstrated that linear probes could extract semantic features from neural networks. This approach has since been validated across numerous concepts, including refusal (Arditi et al., 2024), sentiment (Tigges et al., 2024), spatial relationships and temporal awareness (Gurnee and Tegmark, 2024), and truthfulness (Marks and Tegmark, 2024). At the same time, similar insights have been used for activation-based steering techniques, such as (Turner et al., 2025; Zou et al., 2025; Rimsky et al., 2024). Recent work on sparse autoencoders has demonstrated enabled the decomposition of model activations into interpretable sparse features, including ReLU SAEs (Bricken et al., 2023; Huben et al., 2024), top-k SAEs (Gao et al., 2025), and JumpReLU SAEs (Rajamanoharan et al., 2025). These techniques have also enabled behavioural steering (Templeton et al., 2024).

However, the picture is more nuanced than initially proposed. (Engels et al., 2025) demonstrated that not all language model features are one-dimensionally linear, indicating limitations to current approaches. (Goldman-Wetzler and Turner, 2024) discovered over 800 orthogonal "write code" steering vectors, suggesting that similar behaviours may be encoded through multiple linearly independent representations. This is further supported in (Mack and Turner, 2024a). Similarly, (Bailey et al., 2025) showed that obfuscated activations can bypass safety probes while maintaining harmful capabilities - achieving identical harmful outputs through different activation patterns that evade detection. Despite these limitations, linear methods remain valuable tools for safety analysis and monitoring, complementing black-box approaches while providing mechanistic insights into model behaviour.

There had been prior work applying these insights to safety contexts. (MacDiarmid et al., 2024) found simple probes can detect hidden backdoors in models. (Roger, 2023) trained off-policy "coup probes" which distinguish between regular outputs and synthetic takeover attempts. Similarly, (Goldowsky-Dill et al., 2025b) trained probes to detect deceptive model outputs. (Greenblatt and Shlegeris, 2024) discussed various theoretical considerations related to controlling scheming AIs, citing probes as a specific countermeasure.

3. Methodology

For reproducibility, an implementation of the paper, including data, can be found at [this Github repository](#).

3.1. Model and Data

We conducted our experiments with Llama-3.3-70B-Instruct, loaded at bfloat16 precision on an H200 GPU from Vast.ai. The experiments take approximately 40 GPU hours to complete in total.

For training probes, we constructed several contrastive datasets as detailed in Appendix 6.1. These were either constructed from existing datasets, such as the Situational Awareness Dataset (SAD) stages_oversight dataset (Laine et al., 2024), or synthetic datasets generated by Claude 3.5 Sonnet.

We used several datasets to validate the probes' generalisation. Evaluation prompts were taken from several benchmarks (such as TruthfulQA (Lin et al., 2022)), and deployment prompts were taken from datasets of real user queries or instructions (such as Wildchat (Zhao et al., 2024)). Details can be found in Appendix 6.2.

Lastly, we apply the probes to four safety evaluation datasets designed for deception capabilities. Further details can be found in Appendix 6.3.

We maintained a balanced 50/50 split between options A and B throughout all training datasets to eliminate potential bias. When evaluating probes, we calculated metrics both with and without the SAD data to avoid in-distribution overfitting for the SAD-trained probes.

3.2. Probe Training

Our linear probe training methodology is similar to (Rimsky et al., 2024) and (MacDiarmid et al., 2024). For a given dataset and an N-layer model, we extracted activation vectors from each layer prior to the residual connection, result-

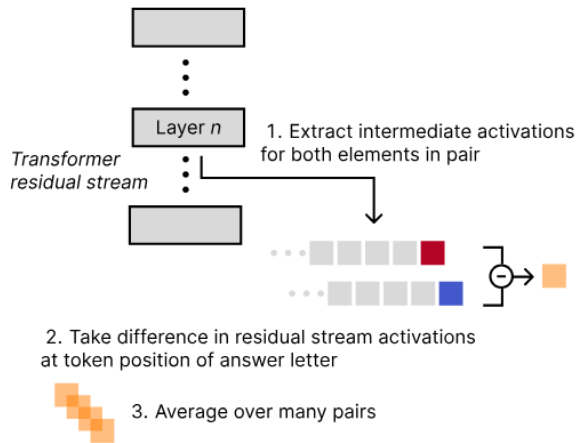


Figure 1. Diagram of linear probe training, adapted from (Rimsky et al., 2024)